

Global Technology Trends Analysis

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EXECUTIVE SUMMARY



- Analyzed global developer survey data to identify current technology usage and future demand trends.
- Findings show strong continuity between present skills and future learning priorities, particularly in programming languages, databases, and cloud platforms.
- HTML/CSS, Bash/Shell, JavaScript, and Python remain core languages, reflecting the growing importance of web development, automation, and DevOps.
- PostgreSQL and Microsoft SQL Server lead both current usage and future demand, while NoSQL and cloud-native databases continue to grow.
- The developer workforce is young, educated, and globally distributed, with increasing emphasis on cloud-first and scalable solutions.
- These insights support informed decisions for developers, educators, and organizations aligning skills with industry needs.



INTRODUCTION



- **Purpose**

To identify key trends in technology usage and future demand among developers.

- **Target Audience**

- Aspiring developers
- Technology professionals
- Educators and curriculum designers
- Employers and hiring managers

- **Value of the Analysis**

- Helps align skill development with industry demand
- Supports data-driven decisions in technology adoption and workforce planning

METHODOLOGY

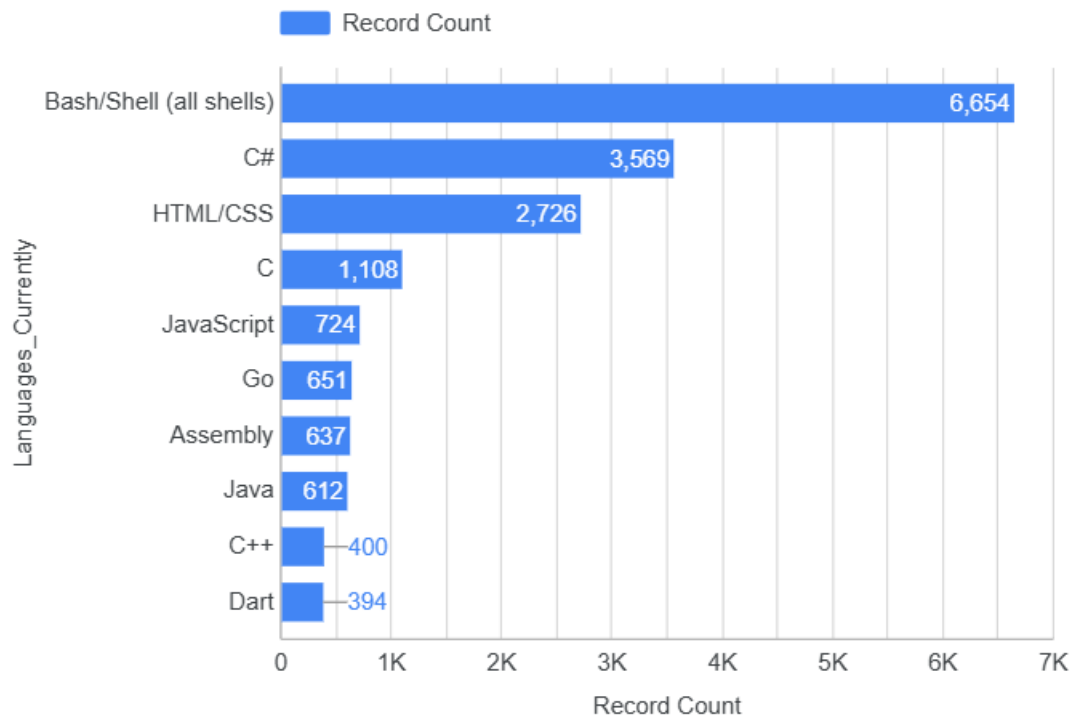


- **Data Source:** Global developer survey dataset
- **Data Collection:** Structured survey responses collected online
- **Key Data Preparation Steps:**
 - Removal of null and inconsistent values
 - Splitting multi-response fields. For example, languages and databases
 - Aggregation of response counts
 - Selection of Top 10 technologies based on frequency
- **Tools Used:**
 - Looker Studio for visualization

PROGRAMMING LANGUAGE TRENDS

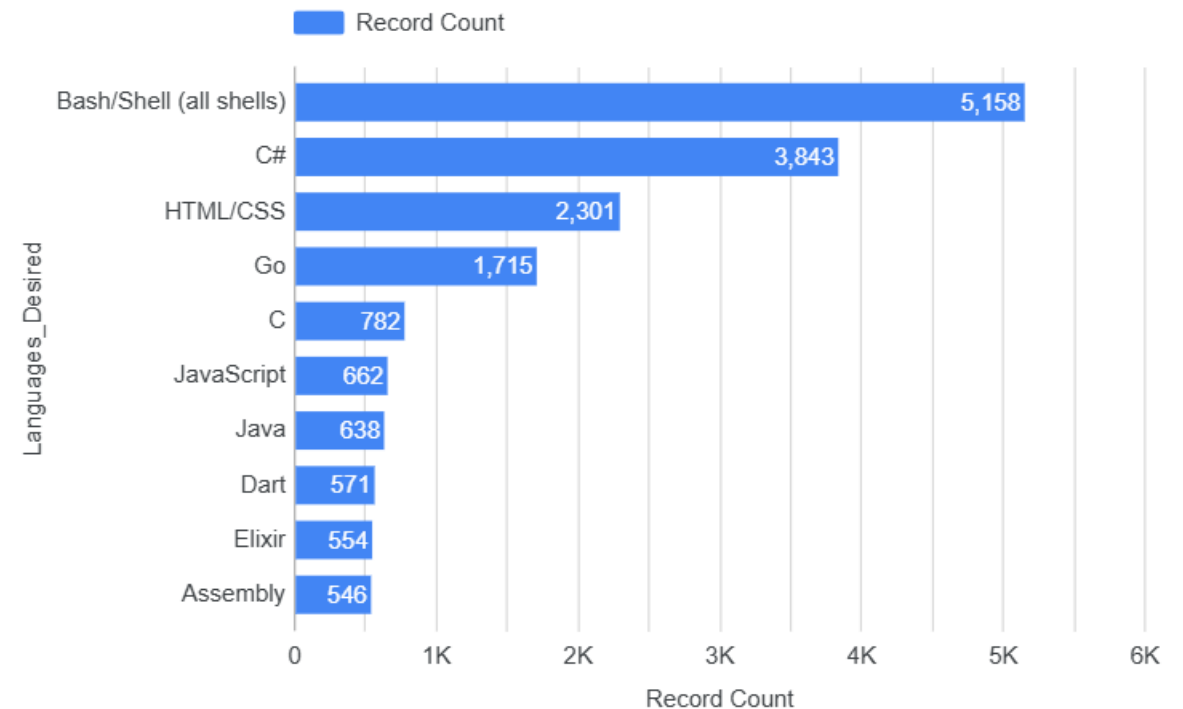
Current Year

Top 10 Languages Used



Next Year

Top 10 Languages Desired Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- Most desired language for next year: Bash/Shell.
- Other highly desired languages: HTML/CSS and C#.
- Java, C++, and Dart are desired less frequently.

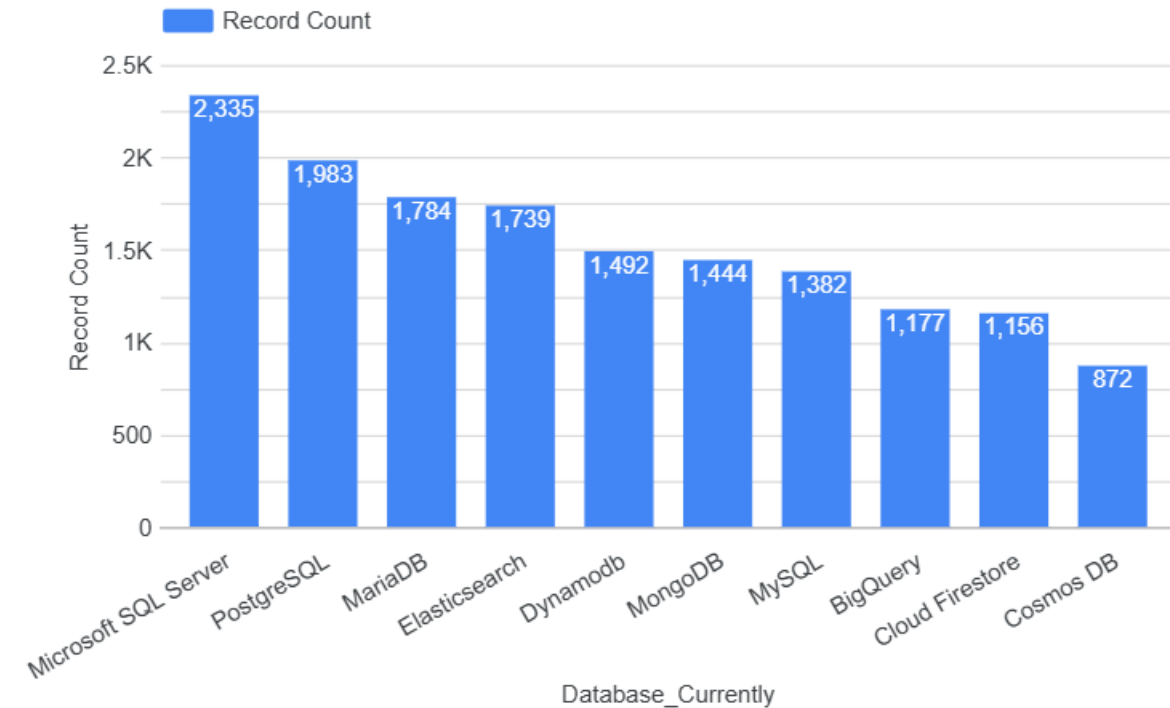
Implications

- Strong interest in DevOps/automation tools (Bash/Shell).
- Enterprise and web development skills (C#, HTML/CSS) remain highly relevant in the job market.
- Growth in Go suggests increasing demand for systems/cloud-native development.
- Traditional languages like Java/C++ are less prioritized for new learning.

DATABASE TRENDS

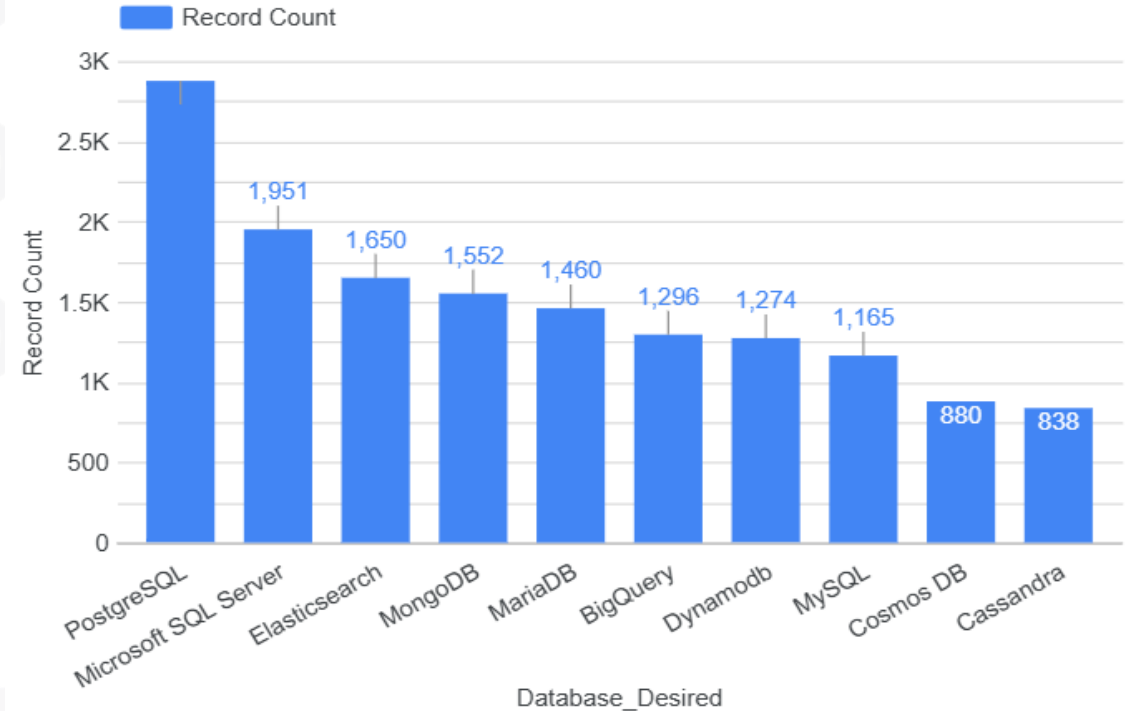
Current Year

Top 10 Databases Used



Next Year

Top 10 Databases Desired Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- Relational databases, particularly Microsoft SQL Server and PostgreSQL, dominate current adoption and future demand.
- NoSQL databases such as Elasticsearch and MongoDB also show strong and sustained usage.
- The increasing preference for cloud-native and scalable database solutions reflects evolving infrastructure and performance needs.

Implications

- Strong demand for SQL skills will persist.
- Hybrid relational NoSQL architectures are becoming standard.
- Cloud database expertise is critical for modern data-driven systems.

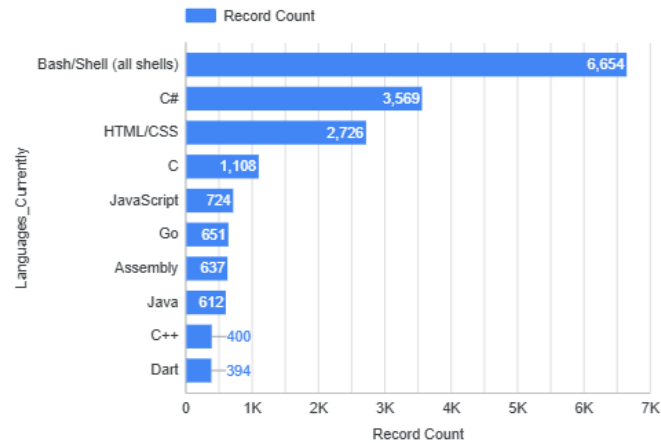
DASHBOARD



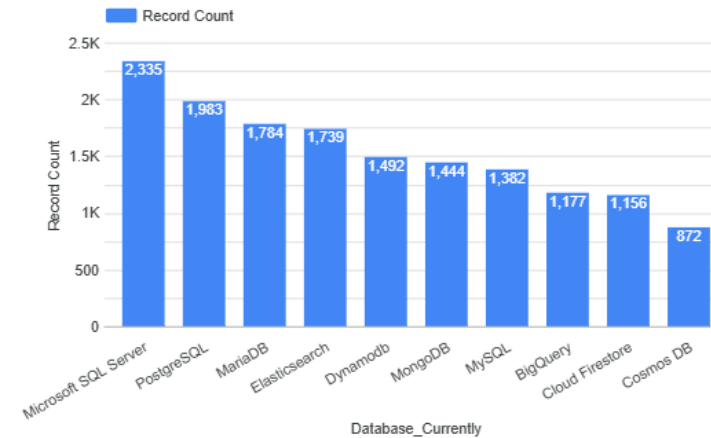
The following dashboard tabs screenshots present summary of current technology usage, anticipated future trends, and respondent demographics.

DASHBOARD TAB 1

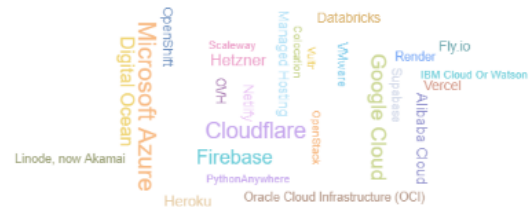
Top 10 Languages Used



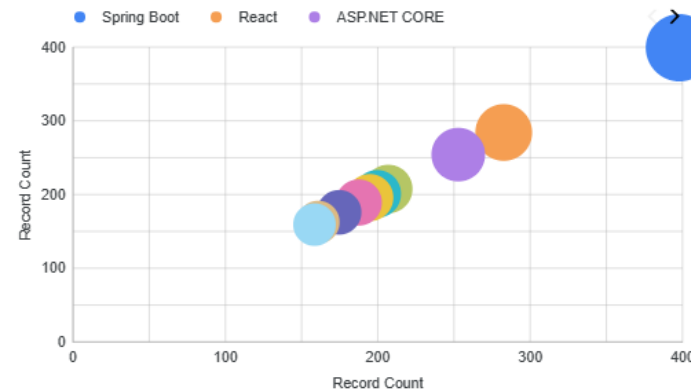
Top 10 Databases Used



Top 10 Platforms Used

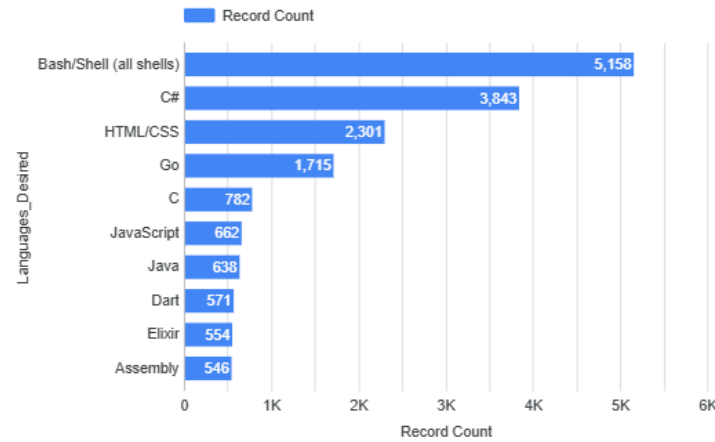


Top 10 Web Frameworks Used

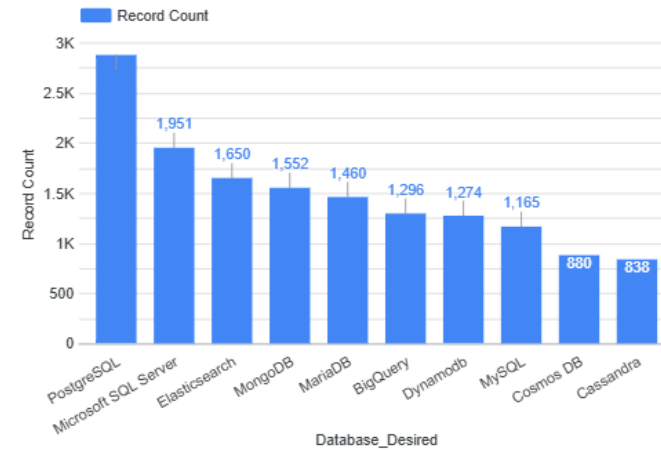


DASHBOARD TAB 2

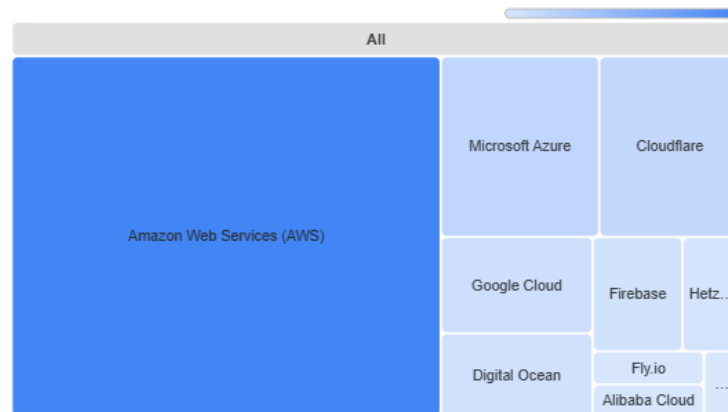
Top 10 Languages Desired Next Year



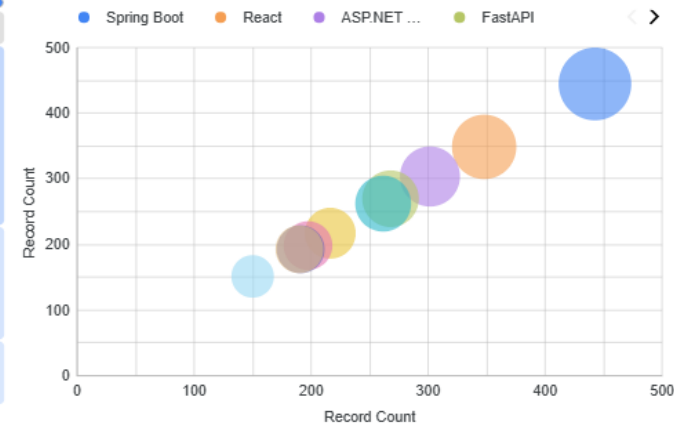
Top 10 Databases Desired Next Year



Top 10 Desired Platforms

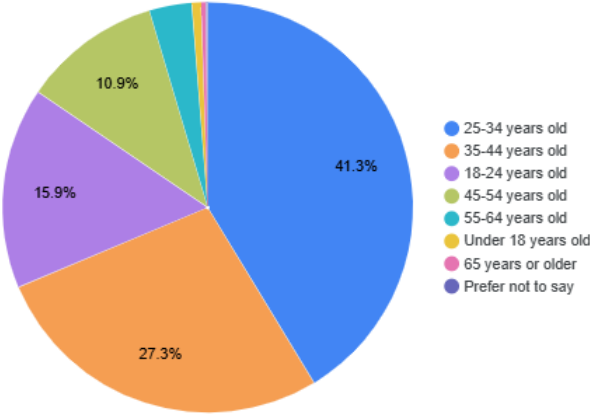


Top 10 Desired Web Frameworks

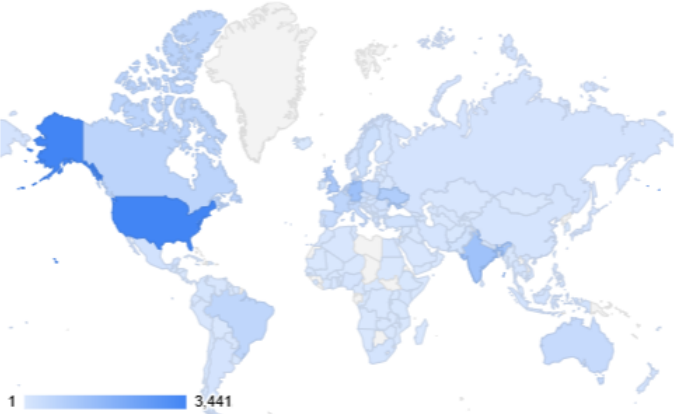


DASHBOARD TAB 3

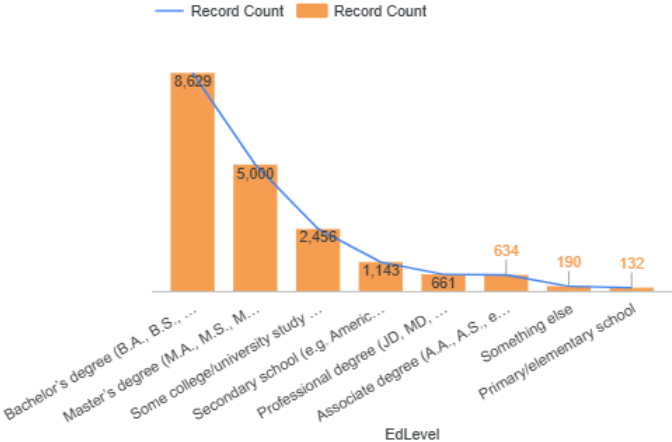
Respondent Classified by Age Group



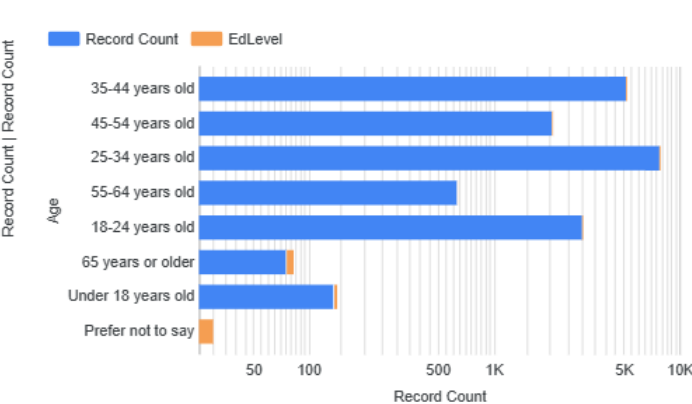
Respondent Count by Country



Respondent Distribution by Education Level



Respondent Classified by Education Level (LogScale)



DISCUSSION



- **Programming Languages**

The dashboard shows Bash/Shell rising sharply in desirability, which reflects the growing importance of DevOps and automation. Meanwhile, Python and JavaScript remain central pillars, but traditional languages like Java and C++ are less prioritized by new learners. This suggests a generational shift toward cloud-native and automation-first skills

- **Databases**

SQL-based systems (PostgreSQL, SQL Server) dominate both current and future usage, but NoSQL options like MongoDB and Elasticsearch are steadily growing. The coexistence of relational and NoSQL databases highlights the industry's move toward hybrid architectures that balance structure with scalability.

- **Platforms & Frameworks**

Cloud platforms (AWS, Azure, Google Cloud) are consistently desired, underscoring the cloud-first development environment. Frameworks like React and Spring Boot remain popular, while FastAPI and Node.js show momentum, pointing to demand for lightweight, modern frameworks.

- **Demographics & Workforce Trends**

The majority of respondents are aged 25–34 with bachelor's or master's degrees, and most are employed full-time. This indicates a relatively mature, professional developer base. The presence of freelancers and contractors also suggests flexible workforce models are becoming more common.



OVERALL FINDINGS & IMPLICATIONS

Findings

Tab 1: Current Technologies Used

- **Languages:** Bash/Shell, C#, HTML/CSS, and Python dominate current usage, reflecting strong demand for automation, enterprise, and web skills.
- **Databases:** SQL Server and PostgreSQL lead adoption, while MongoDB and Elasticsearch highlight the relevance of NoSQL.
- **Platforms:** Cloud platforms (AWS, Azure, Google Cloud) are widely used, confirming the cloud-first environment.
- **Frameworks:** Spring Boot, React, and ASP.NET Core are the most popular, showing balance between enterprise and modern web development.

Implications

Developers must maintain core skills (SQL, Python, JavaScript) while integrating cloud and modern frameworks to stay relevant.



OVERALL FINDINGS & IMPLICATIONS

Findings

Tab 2: Future Desired Technology

- **Languages:** Bash/Shell is the most desired skill, signaling DevOps growth. Go and HTML/CSS also show rising interest.
- **Databases:** PostgreSQL and DynamoDB are highly desired, reflecting hybrid relational + NoSQL adoption.
- **Platforms:** AWS, Azure, and Google Cloud dominate future demand, with Firebase and Cloudflare gaining traction.
- **Frameworks:** Spring Boot, React are remain most desired alongside FastAPI, pointing to lightweight, scalable solutions.

Implications

Future skills will emphasize automation, cloud-native databases, and modern frameworks, requiring continuous learning and adaptation.



OVERALL FINDINGS & IMPLICATIONS

Findings

Tab 3: Demographics

- **Age:** Majority of respondents are 25-34 years old, followed by 18-24, showing a young, dynamic developer base.
- **Country:** Global distribution with strong representation across major tech hubs.
- **Education:** Most hold Bachelor's and Master's degrees, with smaller groups from secondary school or elementary certificates.
- **Employment:** Over two-thirds are full-time employed, but freelancing and contracting remain significant

Implications

The workforce is highly educated and globally distributed, with flexible employment models shaping the industry. Employers must adapt hiring strategies to attract both full-time and freelance talent.



CONCLUSION



Current and Future demand shifts toward automation and cloud

- Bash/Shell, Go, and cloud-native databases/platforms highlight the move to scalable, DevOps-driven ecosystems.

Hybrid architectures are the new standard

- Relational + NoSQL adoption reflects the need for flexibility in modern applications.

Demographics shape the workforce

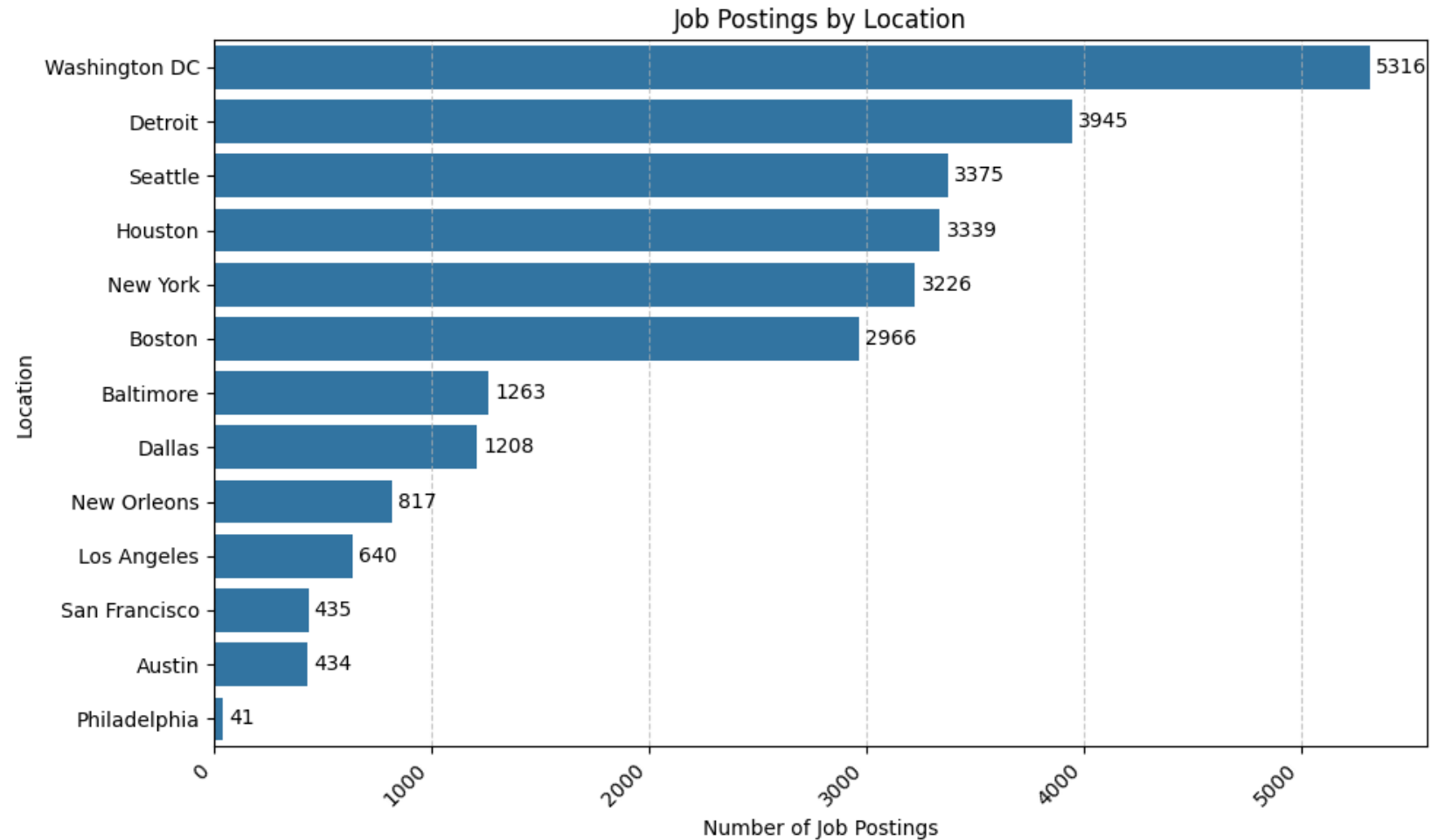
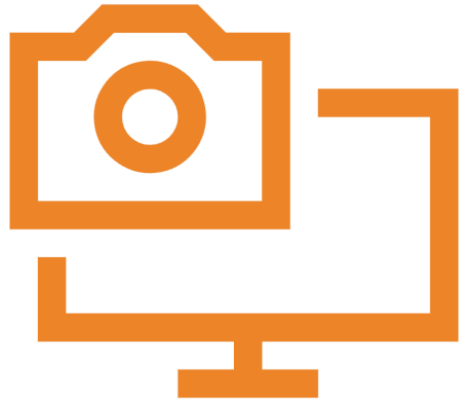
- A young, globally distributed, highly educated developer base drives innovation and requires inclusive hiring strategies.

Continuous learning is critical

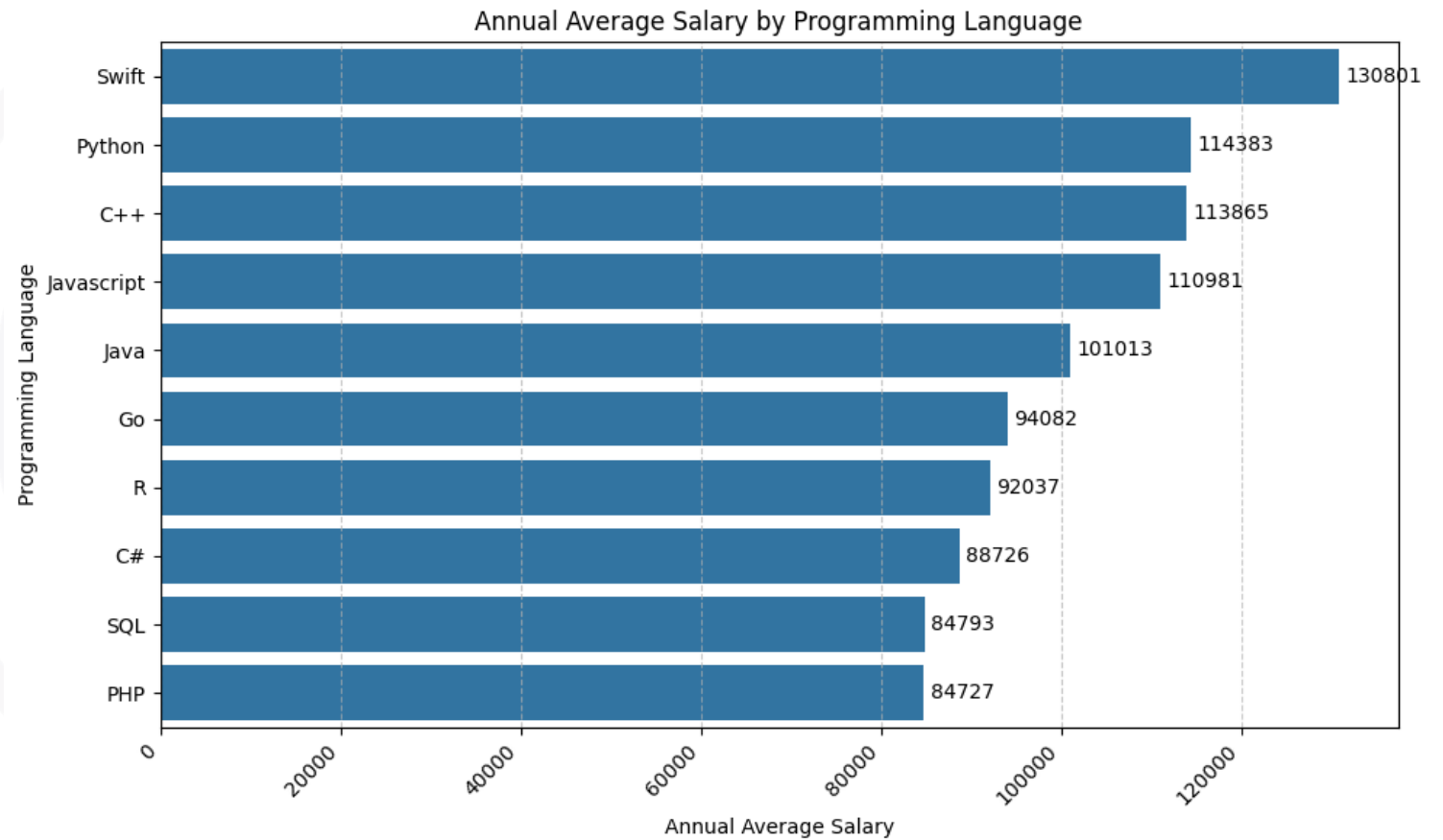
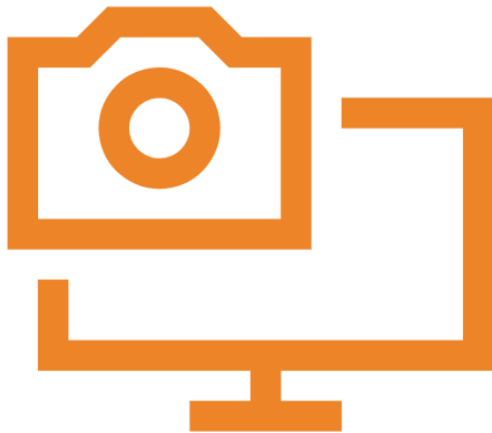
- Developers, educators, and employers must adapt quickly to evolving tools and frameworks to remain competitive.



APPENDIX



APPENDIX



JOB POSTINGS

Findings

- Washington DC leads with the highest number of job postings (5,316), followed by Detroit (3,945) and Seattle (3,375).
- New York (3,226) and Houston (3,339) also show strong demand.
- Philadelphia has the lowest number of postings (41), indicating limited tech hiring activity.
- Mid-tier cities like Boston, Baltimore, and Dallas show moderate demand, while San Francisco and Austin have surprisingly lower counts than expected.

Implications

- Washington DC's dominance suggests strong demand in government tech, cybersecurity, and public sector innovation.
- Detroit's high numbers reflect the growth of tech roles in automotive and manufacturing industries.
- Seattle and Houston indicate robust hiring in cloud services, energy tech, and enterprise software.
- Lower postings in San Francisco and Austin may reflect saturation, remote work shifts, or economic adjustments.
- Job seekers and educators should align training and career strategies with high-demand regions like DC, Detroit, and Seattle.

POPULAR LANGUAGES

Findings

- Swift offers the highest average salary (\$130,801), likely due to its niche demand in iOS development.
- Python (\$114,383) and C++ (\$113,865) follow closely, reflecting their importance in data science, AI, and systems programming.
- JavaScript (\$110,981) remains highly lucrative, supporting its dominance in web development.
- Go (\$94,082) and R (\$92,037) show strong earning potential in cloud-native and statistical computing roles.
- SQL, PHP, and C# fall below the \$90K mark, indicating lower compensation despite widespread usage.

Implications

- High-paying languages like Swift, Python, and C++ suggest strong demand in specialized domains (mobile, AI, systems).
- Web and cloud developers can expect competitive salaries with JavaScript and Go.
- Educators and learners should prioritize Python and JavaScript for broad applicability and strong financial returns.
- SQL and PHP, while foundational, may offer lower salary ceilings, emphasizing the need to pair them with complementary skills. For example, cloud, backend frameworks.
- Career planning should consider both popularity and earning potential to align skill development with financial goals.



SOURCES

- IBM Skills Network. Popular Programming Languages. Retrieved April, 2024 from https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DA0321EN-SkillsNetwork/labs/datasets/Programming_Languages.html.
- Stack Overflow. (2019, February). Annual Developer Survey, 2019. Retrieved April, 2024 from https://insights.stackoverflow.com/survey?_ga=2.177616038.2076724704.1713048265-654320079.1713048265.